

Strategic coexistence theory for evolutionary games

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Abstract

Evolutionary game theory and ecological coexistence theory both seek to predict the outcome of competition between biological entities, be they strategies or species, but the two fields have relied on largely separate approaches. Replicator equations provide a foundation for predicting the outcome of strategic competition, yet they do not explicitly decompose the mechanisms underlying these outcomes. Inspired by modern coexistence theory from community ecology, we develop Strategic Coexistence Theory (SCT), which translates payoff-based invasion conditions into strategic niche differences and fitness differences between any two competing strategies described by classic replicator dynamics. SCT recovers the classic classification of two-strategy games, distinguishing competitive exclusion, coexistence, and priority effects within a shared niche–fitness difference space. Applying SCT to five mechanisms for the evolution of cooperation further reveals that these mechanisms promote cooperation through distinct dynamical routes: kin selection, network reciprocity, and group selection primarily reduce fitness differences, whereas direct and indirect reciprocity destabilize competition and generate priority effects. Finally, applying SCT to microbial public-goods competition shows that nonlinear microbial growth can both stabilize and equalize competition between cooperators and defectors, allowing coexistence. Together, these results show that SCT provides a common language for comparing coexistence mechanisms across evolutionary games.

Significance statement

Introduction

Understanding the mechanisms that promote the evolutionary stability of competing strategies is a fundamental aim in evolutionary biology (Lewontin, 1961; Slobodkin, 1964; Smith and Price, 1973; Taylor and Jonker, 1978). To this end, evolutionary game theory (EGT) and replicator dynamics have provided key foundations, allowing us to predict the outcome of strategic competition through payoff inequalities, invasion criteria, and stability analysis (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014). Recently, there has been

growing interest in characterizing similarities between replicator dynamics and the dynamics of competing species (Gokhale and Traulsen, 2025; Tarnita and Traulsen, 2025; Allesina, 2026), building on the classical mathematical equivalence between replicator equations and generalized Lotka–Volterra models (Hofbauer, 1981; Page and Nowak, 2002). In particular, Tarnita and Traulsen (2025) recently argued that a closer synthesis between ecology and evolutionary game theory can provide new insights into ecological and evolutionary assembly, competition, and coexistence. This renewed perspective between ecological and evolutionary dynamics suggests that evolutionary games may be interpreted not only as payoff-based competition but also as problems of strategic coexistence.

Interestingly, even a simple two-strategy game can yield heterogeneous outcomes that mirror outcomes in community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025): the dominance of a single strategy (prisoner’s dilemma and harmony games), coexistence (hawk–dove games), and bistability (stag-hunt games). Such differences in conditions that promote evolutionary stability and dominance of a single strategy and those that permit strategy coexistence indicate that EGT can be understood from the perspective of coexistence theory (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli, 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). In fact, stable polymorphisms of evolutionary strategies, such as coexistence of cooperators and defectors, are observed in microbial and other biological systems (Gore et al., 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024), but coexistence-theoretic perspective has often remained implicit. Recasting game dynamics as a coexistence problem immediately raises the following question: when two strategies coexist, does the corresponding mechanism stabilize or equalize competition (Chesson, 2000)? While EGT allows us to predict whether two strategies can coexist or not, a framework for quantifying coexistence mechanisms between competing strategies remains elusive.

Modern coexistence theory (MCT) from community ecology presents a particularly useful framework for quantifying coexistence mechanisms (Chesson, 2000; Adler et al., 2007; Godoy et al., 2014; Godoy and Levine, 2014; Kraft et al., 2015; Adler et al., 2026). MCT predicts that coexistence of two competing entities, be they species or strategies, requires stabilizing niche differences to overcome average fitness differences (Chesson, 2000; Adler et al., 2007; Kraft et al., 2015). In community ecology, stabilizing niche differences represent differences in factors that limit the growth of competing species, causing each species to limit itself more than it limits its competitor (Chesson, 2000). In EGT, a strategy’s niche can be interpreted as strategic environments defined by the composition of competing strategies and the resulting interactions (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025). Thus, stabilizing niche differences arise when each strategy is favored by a different strategic environment, such that the spread of each strategy creates conditions that limit its own relative growth and favor the recovery of its competitor. In contrast, fitness differences represent differences in the ability of

each entity to compete under the limiting environments created by its competitor (Chesson, 2000). This differs from payoff differences in EGT, which depend on the underlying strategy frequency and therefore combine both stabilizing effects caused by changes in the strategic environment and residual fitness differences that favor one strategy over another (Taylor and Jonker, 1978; Cressman and Tao, 2014; Tarnita and Traulsen, 2025).

So far, MCT has been primarily utilized for understanding species coexistence, but recent work showed that MCT can be extended beyond species competition, especially for studying pathogen strain competition, allowing one to directly quantify niche and fitness differences of competing entities, be they species or strains (Sieben et al., 2022; Park et al., 2024, 2026). Inspired by this effort, we present Strategic Coexistence Theory (SCT) for predicting coexistence and exclusion of competing strategies in EGT by quantifying strategic niche and fitness differences. We begin by introducing SCT, which can be applied to any generic system of EGT described by replicator equations. We show that SCT can recover outcomes from classic, two-strategy games, providing a reinterpretation of classic results from coexistence-theoretic perspective. We then apply SCT to comparing five mechanisms for the evolution of cooperation, revealing that direct and indirect reciprocity destabilize competition between cooperators and defectors, thus pushing the system towards priority effect regime. Finally, we consider a case study of microbial cooperation, where nonlinear microbial growth can both stabilize and equalize competition between cooperation and defection, allowing coexistence of strategies.

Strategic coexistence theory

Here, we present a brief derivation of strategic coexistence theory (SCT) (Figure 1). To do so, we begin with a classic replicator equation, which allows us to model changes in the frequency of competing strategies based on their payoffs (Schuster and Sigmund, 1983; Hofbauer and Sigmund, 1998; Cressman, 2003; Cressman and Tao, 2014):

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (1)$$

where x_i represents the frequency of strategy i , $\pi_i(\mathbf{x})$ represents the payoff of strategy i for a given strategy distribution $\mathbf{x}$, and $\bar{\pi}(\mathbf{x}) = \sum_i \pi_i(\mathbf{x})x_i$ represents the average payoff. Since strategy frequencies must sum to 1 (i.e., $\sum_i x_i = 1$), a single-strategy equilibrium dominated by strategy i can be described by a unit vector $\mathbf{e}_i$, where all elements are equal to zero except for the i -th element. We use these single-strategy equilibria to define niche and fitness differences between two competing strategies i and j .

Because payoffs in many games can be negative, payoff ratios can be difficult to interpret. Thus, we work on a positive multiplicative scale by exponentiating the payoff terms:

$$W_i(\mathbf{x}) = \exp(\pi_i(\mathbf{x})). \quad (2)$$

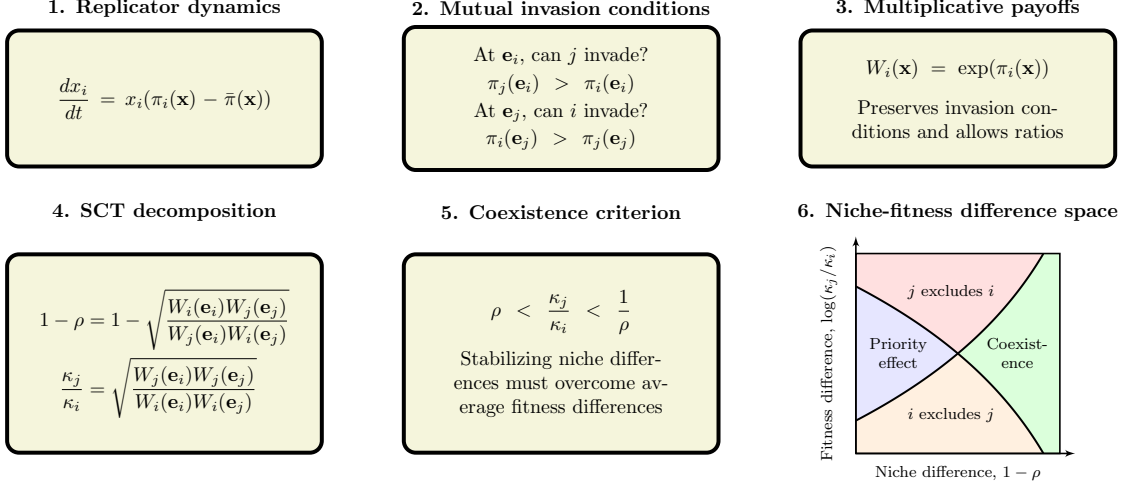


Figure 1: **A schematic diagram summarizing strategic coexistence theory.** Strategic coexistence theory decomposes pairwise replicator dynamics into stabilizing and equalizing mechanisms. Mutual invasion conditions can be rewritten on a multiplicative payoff scale, allowing us to derive strategic niche difference $1 - \rho$ and fitness difference κ_j/κ_i based on multiplicative payoff ratios. Then, the inequality $\rho < \kappa_j/\kappa_i < 1/\rho$ determines mutual invasion.

118 This exponential transformation is convenient because it preserves the payoff order-
 119 ing and does not alter the invasion conditions. Therefore, this allows us to compare
 120 strategic environments through ratios, as in modern coexistence theory. For conven-
 121 ience, we refer to $W_i(\mathbf{x})$ as the multiplicative payoff of strategy i .

122 First, we define the niche overlap ρ between strategies i and j as the ratio between
 123 the geometric mean of each strategy's multiplicative payoff in the strategic environ-
 124 nment that it creates when common, $\sqrt{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}$, and the geometric mean of
 125 each strategy's multiplicative payoff in the environment generated by its competitor,
 126 $\sqrt{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}$. The corresponding niche difference is then given by

$$1 - \rho = 1 - \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (3)$$

127 When $1 - \rho > 0$, each strategy performs better, on average, in the environment
 128 generated by its competitor than in the environment generated by itself. In other
 129 words, intraspecific competition is stronger than interspecific competition. Second,
 130 we define the fitness difference as the ratio between the geometric means of the
 131 multiplicative payoff of each strategy across the two single-strategy environments:

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}. \quad (4)$$

132 In other words, strategy j has an average competitive advantage over strategy i
 133 ($\kappa_j/\kappa_i > 1$) when it performs better, on average, across the environments generated
 134 by itself and its competitor.

135 Together, for any two strategies described by the classic replicator equation, mu-
 136 tual invasion requires stabilizing niche differences to overcome average fitness differ-
 137 ences. Specifically, the condition for mutual invasion can be rewritten as (Chesson,
 138 2000):

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (5)$$

139 This inequality applies to any games described by the classic replicator equation
 140 because ρ and κ_j/κ_i can be calculated directly from ratios of multiplicative pay-
 141 offs. This decomposition not only allows us to understand how different mechanisms
 142 contribute to the coexistence or exclusion of competing strategies but also provides
 143 shared axes for comparing different games. Detailed derivations are provided in
 144 Supplementary Text S1.

145 We note that the absolute magnitudes of niche and fitness differences are sensitive
 146 to payoff scaling because they are defined on an exponentiated payoff scale. However,
 147 the qualitative outcome of strategic competition under SCT is preserved because the
 148 exponential transformation does not alter invasion conditions.

149 Classic two-strategy games

150 As an illustrative example, we begin with classic two-strategy games (Hauert, 2002;
 151 Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025), which can be
 152 described by the following payoff matrix (Figure 2A):

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (6)$$

153 Here, the first and second rows represent strategies 1 and 2, respectively, and the
 154 columns represent the strategy of the interacting opponent. Writing x_1 to denote
 155 the frequency of strategy 1, the expected payoffs are

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (7)$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (8)$$

156 Thus, the dynamics can be written as

$$\frac{dx_1}{dt} = x_1(1 - x_1) [(R - T)x_1 + (S - P)(1 - x_1)]. \quad (9)$$

157 The signs of $T - R$ and $S - P$ determine whether each strategy can invade the
 158 other when rare. Specifically, strategy 2 can invade strategy 1 when $T > R$, whereas
 159 strategy 1 can invade strategy 2 when $S > P$. These two invasion conditions partition

160 the game space into four regimes (Figure 2B): prisoner’s dilemma ($T > R > P >$
 161 S), hawk–dove game ($T > R, S > P$), stag–hunt game ($R > T, P > S$), and
 162 harmony game ($R > T, S > P$). In standard EGT, these outcomes are obtained by
 163 analyzing the stability of equilibrium conditions. For simple games, this analysis is
 164 straightforward.

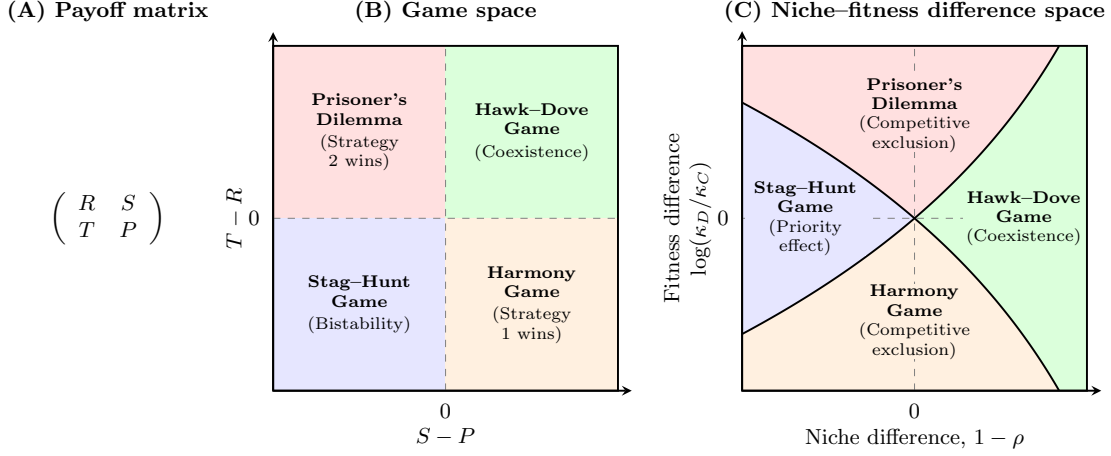


Figure 2: **Coexistence-theoretic interpretation of classic two-strategy games.** (A) General payoff matrix for a two-strategy game, where rows represent the focal strategy and columns represent the strategy of the interacting opponent. (B) Classification of four classic games based on the payoff differences $T - R$ and $S - P$. Prisoner’s dilemma and harmony games lead to the dominance of one strategy; hawk–dove games allow coexistence; and stag–hunt games result in bistability. (C) Mapping of the same four games onto niche–fitness difference space. Hawk–dove games correspond to the coexistence regime, where stabilizing niche differences overcome fitness differences. Prisoner’s dilemma and harmony games correspond to competitive-exclusion regimes, whereas stag–hunt games correspond to the priority-effect regime. $R > P$ is assumed throughout, but this does not affect coexistence outcome in any way.

165 A coexistence-theoretic approach recovers the same classification but provides
 166 a different interpretation (Figure 2C). For a two-strategy game determined by this
 167 payoff matrix, the niche difference and fitness difference are given by the following
 168 set of equations (Supplementary Text S2):

$$1 - \rho = 1 - \exp\left(\frac{R + P - S - T}{2}\right), \quad (10)$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (11)$$

169 As shown above, mutual invasion requires $\rho < \kappa_2/\kappa_1 < 1/\rho$, which is equivalent to
 170 $T > R$ and $S > P$. This corresponds to the hawk–dove game, in which each strategy

171 can invade its competitor when rare and the two strategies coexist. The remaining
172 games also have natural interpretations in niche–fitness difference space. Prisoner’s
173 dilemma and harmony games represent competitive exclusion regimes, where one
174 strategy has a fitness advantage that exceeds niche differences. In contrast, stag–hunt
175 games represent a priority-effect regime, where neither strategy can invade the other
176 and the outcome depends on initial conditions. Therefore, coexistence theory not
177 only allows us to recover the classic results from two-strategy games but also provides
178 new interpretations for these results in terms of niche and fitness differences. Building
179 on this, we next apply this framework to comparing core mechanisms of cooperation.

180 Comparisons of mechanisms for the evolution of co- 181 operation

182 Even though cooperative behavior can be found across a wide range of biological
183 systems, a simple prisoner’s dilemma shows that cooperation is not an evolutionarily
184 stable strategy (Axelrod and Hamilton, 1981; May, 1987). Thus, understanding
185 mechanisms that promote the evolution of cooperation has remained a central goal
186 in evolutionary biology (West et al., 2007; Fletcher and Doebeli, 2008; Clutton-Brock,
187 2009; Kingma et al., 2014; Rakoff-Nahoum et al., 2016; Hilbe et al., 2018; Apicella
188 and Silk, 2019; West et al., 2021; Lee et al., 2022; Palmer and Foster, 2022; Michel-
189 Mata et al., 2024). Here, we compare five classic mechanisms for the evolution of
190 cooperation using SCT (Nowak, 2006) and ask how they differ from the perspective
191 of SCT.

192 Specifically, we consider the following mechanisms: kin selection, direct reci-
193 procity, indirect reciprocity, network reciprocity, and group selection (Figure 3A).
194 Kin selection favors cooperation when two interacting individuals are genetic rela-
195 tives, where the genetic relatedness r must be greater than the cost-to-benefit ratio:
196 $r > c/b$ (Hamilton, 1964). Direct reciprocity favors cooperation when current co-
197 operation may promote future cooperation by the same individual through repeated
198 encounters (Trivers, 1971; Axelrod and Hamilton, 1981). In this case, the probability
199 of repeated interaction (i.e., “next round”) must be greater than the cost-to-benefit
200 ratio: $w > c/b$. Indirect reciprocity favors cooperation when current cooperation
201 may promote future cooperation by different individuals through reputation (Nowak
202 and Sigmund, 1998; Wedekind and Milinski, 2000; Nowak and Sigmund, 2005). Thus,
203 the social acquaintanceship (i.e., the probability of knowing someone’s reputation)
204 must be greater than the cost-to-benefit ratio: $q > c/b$. Network reciprocity favors
205 cooperation via clustering of cooperators, requiring the benefit-to-cost ratio to be
206 greater than the average number of neighbors: $b/c > k$ (Nowak and May, 1992;
207 Lieberman et al., 2005). Finally, group selection favors cooperation via multilevel
208 selection, where groups with more cooperators exhibit faster growth at the group
209 level even if defectors reproduce faster at the individual level (Traulsen and Nowak,
210 2006). This requires $b/c > 1 + (n/m)$, where n is the maximum group size and

211 m is the number of groups. All of these mechanisms can be described by a simple
 212 2×2 payoff matrix (Nowak, 2006), from which the corresponding niche and fitness
 213 differences can be computed using Eq. (10) and Eq. (11). Details on niche and fitness
 214 difference calculations are presented in Supplementary Text S3.

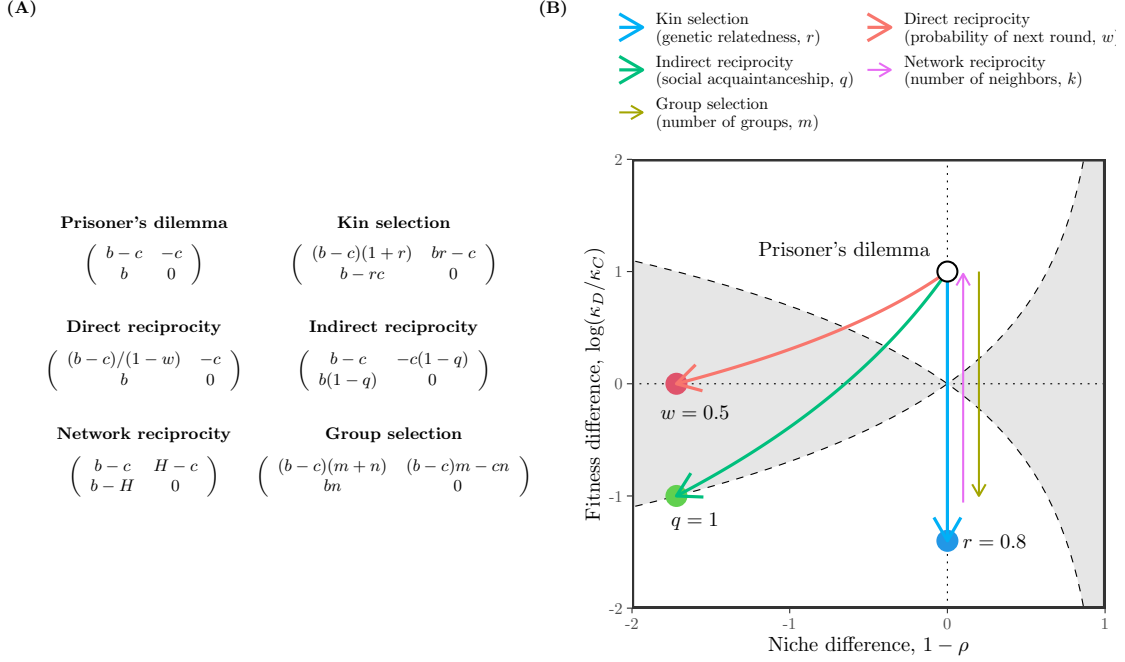


Figure 3: Comparisons of five mechanisms for the evolution of cooperation using SCT. (A) Payoff matrices for the baseline scenario (prisoner's dilemma) and five mechanisms for the evolution of cooperation (Nowak, 2006). Prisoner's dilemma is characterized by the benefit for the recipient b and the cost for the donor c . Network reciprocity is parameterized based on $H = [(b-c)k - 2c]/[(k+1)(k-2)]$. Parameters for all other mechanisms are described in panel B. (B) Changes in niche and fitness differences driven by each mechanism. Each arrow represents the direction of change as we increase the corresponding parameter, indicated in figure legends. Note that kin selection, network reciprocity, and group selection all result in changes in fitness difference without causing any changes in niche difference. Therefore, we indicate the direction of change for network reciprocity and group selection separately to avoid overlapping arrows. For simplicity, we focus on the effect of number of groups m for group selection; increasing the number of maximum group size n has the opposite effect (Supplementary Text S3).

215 First, the prisoner's dilemma provides a baseline scenario, where defection is
 216 evolutionarily stable (Figure 3A). From the perspective of SCT, there is complete
 217 niche overlap, and therefore defection competitively excludes cooperation (Figure
 218 3B). Building on this, SCT shows that five mechanisms promote evolutionary sta-

219 bility of cooperation via different dynamical routes (Figure 3B). For example, kin
 220 selection allows the fitness of cooperation to increase as genetic relatedness r increases
 221 (Figure 3B). However, it does not affect the niche difference between cooperation and
 222 defection. Similarly, network reciprocity and group selection modulate the fitness of
 223 cooperation without causing any changes in strategic niches. In contrast, direct
 224 and indirect reciprocity destabilize competition while increasing the fitness of coop-
 225 eration, thereby pushing the competition outcome towards a priority effect regime.
 226 Equivalently, these two mechanisms promote bistability of defection and cooperation.

227 SCT further reveals another difference between direct and indirect reciprocity.
 228 In particular, direct reciprocity is unable to generate sufficient fitness difference to
 229 competitively exclude defection even when the probability of next round w reaches
 230 1 (Supplementary Text S3). In contrast, indirect reciprocity is able to generate
 231 sufficient fitness difference to push the system towards the boundary between priority-
 232 effect regime and competitive-exclusion regime as social acquaintanceship q reaches 1,
 233 indicating that an additional mechanism may be able to promote complete dominance
 234 of cooperation (Supplementary Text S3).

235 We note that none of these mechanisms allow coexistence of cooperation and de-
 236 fection. In many biological systems, such coexistence is often observed (Gore et al.,
 237 2009; Cordero et al., 2012; Kocher et al., 2018; Pollak et al., 2021; Lin et al., 2024).
 238 Therefore, we move on to the final example, applying SCT to microbial coopera-
 239 tion, where nonlinear benefits and environmental context can promote coexistence
 240 of cooperators and defectors.

241 Microbial cooperation and public goods

242 Here, we consider a simple, phenomenological model of competition between cooper-
 243 ating and cheating yeast in a cooperative sucrose-metabolism system, where invertase
 244 production by cooperators allows cheaters to utilize glucose, a public good released
 245 through sucrose hydrolysis (Carlson and Botstein, 1982; Dickinson, 1998; Gore et al.,
 246 2009). Specifically, invertase production by cooperators incurs a cost c and gener-
 247 ates a total benefit of 1 that is captured privately with efficiency ϵ . Assuming a
 248 linear relationship between glucose availability and microbial growth, the payoffs for
 249 cooperators π_C and cheaters (defectors) π_D can then be written as

$$\pi_C(f) = \epsilon + f(1 - \epsilon) - c, \quad (12)$$

$$\pi_D(f) = f(1 - \epsilon), \quad (13)$$

250 where the use of glucose depends on the fraction of cooperators f . The dynamics of
 251 cooperators and defectors are described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\pi_C(f) - \pi_D(f)] \quad (14)$$

$$= f(1 - f)(\epsilon - c) \quad (15)$$

For this linear growth model, the payoff difference is fixed to $\epsilon - c$ independent of f (Figure 4A). Therefore, cooperation is favored when efficiency is greater than the cost of cooperation, $\epsilon > c$, whereas defection is favored when the cost is greater than efficiency, $c > \epsilon$ (Gore et al., 2009).

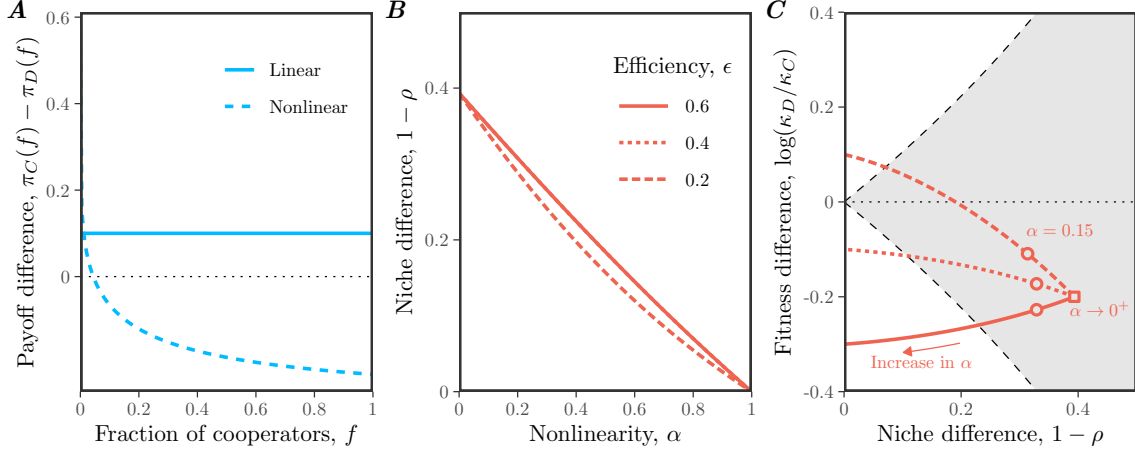


Figure 4: Effects of nonlinear microbial growth on coexistence of cooperation and defection. (A) Payoff differences between cooperation and defection for linear and nonlinear models of microbial growth. In panel A, we assumed $\epsilon = 0.4$, $c = 0.3$, and $\alpha = 0.15$ for illustrative purposes. (B) Effect of nonlinearity α on niche difference $1 - \rho$. Each line assumes a different value of efficiency ϵ . Note that lines for $\epsilon = 0.6$ and $\epsilon = 0.4$ are nearly overlapping and are indistinguishable. (C) Effect of nonlinearity α on both niche difference $1 - \rho$ and fitness difference $\log(\kappa_D/\kappa_C)$. The gray shaded area represents the coexistence regime. The square represents the condition when $\alpha \rightarrow 0^+$. The circles represent the condition when $\alpha = 0.15$, which correspond to the experimental relationship between glucose and microbial growth (Gore et al., 2009). In panels B and C, we assumed $c = 0.3$, while allowing ϵ and α to vary.

Our framework allows us to reinterpret this result from the perspective of coexistence theory. In particular, the two strategies have no niche difference, corresponding to complete niche overlap, $\rho = 1$. In contrast, the fitness difference is determined by the balance between private efficiency and the cost of cooperation:

$$\frac{\kappa_D}{\kappa_C} = \exp(c - \epsilon). \quad (16)$$

Therefore, this model always predicts competitive exclusion, except in the neutral case $c = \epsilon$. The outcome is determined entirely by the fitness difference between cooperators and defectors, rather than by stabilizing niche differences. This example resembles the prisoner's dilemma discussed in the previous example (Figure 3).

In real biological systems, the effect of glucose on microbial growth follows a

265 nonlinear relationship:

$$\pi_C(f) = [\epsilon + f(1 - \epsilon)]^\alpha - c, \quad (17)$$

$$\pi_D(f) = [f(1 - \epsilon)]^\alpha, \quad (18)$$

266 where $\alpha = 0.15$ reflects the experimental relationship between glucose and microbial
 267 growth (Gore et al., 2009). Then, the dynamics of cooperators and defectors are
 268 described by the following replicator equation:

$$\frac{df}{dt} = f(1 - f) [\{\epsilon + f(1 - \epsilon)\}^\alpha - c - \{f(1 - \epsilon)\}^\alpha]. \quad (19)$$

269 Under this nonlinear assumption, the payoff difference decreases with the cooperator
 270 fraction f , which can promote the coexistence of cooperators and defectors (Figure
 271 4A): cooperators have a competitive advantage when rare, whereas defectors have a
 272 competitive advantage when cooperators are abundant.

273 From a coexistence-theoretic perspective, this means that nonlinear growth can
 274 generate sufficient stabilizing niche difference to overcome the fitness difference. In
 275 particular, the niche and fitness differences for this model correspond to:

$$1 - \rho = 1 - \exp\left(\frac{1 - \epsilon^\alpha - (1 - \epsilon)^\alpha}{2}\right), \quad (20)$$

$$\frac{\kappa_D}{\kappa_C} = \exp\left(c + \frac{(1 - \epsilon)^\alpha - 1 - \epsilon^\alpha}{2}\right). \quad (21)$$

276 These expressions clarify the separate roles of cost, efficiency, and nonlinearity. For
 277 example, the cost of cooperation does not affect niche difference and only affects
 278 fitness difference. Instead, niche difference is generated by the concave relationship
 279 between glucose availability and microbial growth, which increases from 0 to $1 - \exp(-1/2)$
 280 as α goes from 1 to 0 (Figure 4B,C). In contrast, as $\alpha \rightarrow 0^+$, the fitness
 281 difference approaches

$$\log(\kappa_D/\kappa_C) \rightarrow c - \frac{1}{2}, \quad (22)$$

282 which no longer depends on the private capture efficiency ϵ (Figure 4C). Thus, strong
 283 concavity not only generates stabilizing niche differences, but also acts as an equal-
 284 izing mechanism (Figure 4C). Such stabilizing mechanisms are key ingredients for
 285 coexistence of cooperators and defectors.

286 Discussion

287 Standard evolutionary game theory and replicator equations provide powerful tools
 288 for predicting the outcome of strategy competition, including strategic dominance,
 289 coexistence, and bistability (Motro, 1991; Doebeli et al., 2004; Hauert and Doebeli,
 290 2004; Doebeli and Hauert, 2005; Gore et al., 2009; Souza et al., 2009; Archetti and

Scheuring, 2011; Steidinger and Bever, 2014; Tarnita and Traulsen, 2025). However, these outcomes are typically understood by analyzing payoff differences within individual games, which makes it difficult to compare coexistence mechanisms across models. Building on Chesson’s Modern Coexistence Theory from community ecology (Chesson, 2000), we developed a strategic coexistence theory (SCT) that allows us to decompose strategy competition in terms of stabilizing niche differences and fitness differences. This decomposition provides a unifying framework for comparing different games from the perspective of coexistence theory, allowing us to identify stabilizing and equalizing mechanisms for strategy competition.

To demonstrate the utility of SCT, we considered three case studies. First, the analysis of classic two-strategy games showed that SCT recovers the original classification for these classic games, highlighting the similarities between evolutionary games and community ecology (Hauert, 2002; Szabó and Fath, 2007; Cooney, 2020; Tarnita and Traulsen, 2025). Then, we compared different mechanisms for the evolution of cooperation, which showed that each mechanism promotes cooperation via different dynamical routes (Nowak, 2006). Kin selection, network reciprocity, and group selection primarily affect the fitness difference between cooperation and defection, whereas direct and indirect reciprocity can destabilize competition and generate priority effects. Finally, our analysis of microbial cooperation showed that coexistence between cooperators and defectors requires more than reducing the fitness cost of cooperation: nonlinear benefits can generate stabilizing niche differences while also reducing fitness asymmetry (Gore et al., 2009). Taken together, these examples show that SCT can provide new insights into ecological underpinnings of strategy competition.

There are several limitations to SCT. First, similar to MCT from community ecology, SCT only allows for comparisons of pairwise strategies and therefore cannot determine the outcomes of games with more than two strategies (Chesson, 2000). Second, we focused on classic replicator equations, but Tarnita and Traulsen (2025) recently pointed out that these classic formulations implicitly ignore differences in intrinsic growth rates. In such cases, replicator equations may fail to predict the outcome of competition (Tarnita and Traulsen, 2025), and a full Lotka-Volterra model is needed to describe the abundances, rather than frequencies, of strategies. Thus, SCT inherits the same assumptions as the classic replicator framework: it is most appropriate when strategy competition can be described in terms of relative frequencies alone (Tarnita and Traulsen, 2025). When strategies differ intrinsically in growth, MCT may be required instead to tease apart mechanisms of strategy coexistence. Finally, we have focused on simple examples to illustrate the utility of SCT, but many games involve additional complexities, including population structure (Nowak et al., 2010), stochasticity (Wang et al., 2025; Bodin et al., 2026), and explicit ecological feedback (Tarnita and Traulsen, 2025). Extending SCT to these settings will be necessary to determine how broadly stabilizing and equalizing mechanisms can be compared across evolutionary games.

In conclusion, SCT provides a bridge between evolutionary game theory and

community ecology, allowing strategy competition to be understood from the perspective of coexistence theory. Therefore, rather than replacing existing EGT or replicator-dynamics frameworks, SCT provides a complementary tool for teasing apart the coexistence mechanisms underlying evolutionary games. More broadly, our study builds on recent efforts to extend MCT beyond species competition (Park et al., 2024, 2026), providing a foundation for building unifying coexistence theory for competing entities, including species, pathogen strains, and behavioral strategies.

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Competing interests

The authors declare no competing interest.

Data availability

All data and code are stored in a publicly available GitHub repository (<https://github.com/parklab-snu/coexistence-strategy>).

353 Supplementary Text

354 S1 Derivation of the strategic coexistence theory

355 To derive strategic coexistence theory (SCT), we begin with replicator equations:

$$\frac{dx_i}{dt} = x_i(\pi_i(\mathbf{x}) - \bar{\pi}(\mathbf{x})), \quad (\text{S1})$$

356 where x_i represents the frequency of strategy i , π_i represents the payoff of strategy
 357 i , and $\bar{\pi}$ represents the average payoff. Then, the mutual invasion condition for two
 358 competing strategies i and j can be written as

$$\pi_j(\mathbf{e}_j) < \pi_i(\mathbf{e}_j), \quad (\text{S2})$$

$$\pi_i(\mathbf{e}_i) < \pi_j(\mathbf{e}_i). \quad (\text{S3})$$

359 These inequalities will be preserved when we exponentiate the payoffs, $W_i(\mathbf{x}) =$
 360 $\exp(\pi_i(\mathbf{x}))$:

$$W_j(\mathbf{e}_j) < W_i(\mathbf{e}_j), \quad (\text{S4})$$

$$W_i(\mathbf{e}_i) < W_j(\mathbf{e}_i). \quad (\text{S5})$$

361 Rearranging, we obtain the following inequality:

$$\frac{W_i(\mathbf{e}_i)}{W_j(\mathbf{e}_i)} < 1 < \frac{W_i(\mathbf{e}_j)}{W_j(\mathbf{e}_j)}. \quad (\text{S6})$$

362 By multiplying $\sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}$ on both sides, we obtain:

$$\sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}} < \sqrt{\frac{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}} \quad (\text{S7})$$

363 Thus, by defining niche overlap ρ and fitness difference κ_j/κ_i as follows,

$$\rho = \sqrt{\frac{W_i(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_j(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S8})$$

$$\frac{\kappa_j}{\kappa_i} = \sqrt{\frac{W_j(\mathbf{e}_i)W_j(\mathbf{e}_j)}{W_i(\mathbf{e}_i)W_i(\mathbf{e}_j)}}, \quad (\text{S9})$$

364 we obtain the following inequality that defines conditions for mutual invasion and
 365 long-term coexistence of two competing strategies:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S10})$$

366 S2 Applications to classic two-strategy games

367 Consider a two-strategy game described by the following payoff matrix:

$$\begin{pmatrix} R & S \\ T & P \end{pmatrix}. \quad (\text{S11})$$

368 There are four classic games that we can define from this simple payoff matrix:
 369 prisoner's dilemma ($T > R > P > S$), hawk-dove game ($T > R, S > P$), stag-hunt
 370 game ($R > T, P > S$), and harmony game ($R > T, S > P$). Here, we show that
 371 each game corresponds to a distinct region in the niche-fitness difference space under
 372 SCT.

373 To do so, we first begin by deriving expressions for niche overlap ρ and fitness
 374 difference κ_j/κ_i . Writing x_1 to denote the frequency of strategy 1, the expected
 375 payoff for each strategy corresponds to:

$$\pi_1(\mathbf{x}) = Rx_1 + S(1 - x_1), \quad (\text{S12})$$

$$\pi_2(\mathbf{x}) = Tx_1 + P(1 - x_1). \quad (\text{S13})$$

376 Then, multiplicative payoffs for each strategy when rare can be written as:

$$W_1(1) = \exp(R) \quad (\text{S14})$$

$$W_1(0) = \exp(S) \quad (\text{S15})$$

$$W_2(1) = \exp(T) \quad (\text{S16})$$

$$W_2(0) = \exp(P) \quad (\text{S17})$$

377 Therefore, we have:

$$\rho = \exp\left(\frac{R + P - S - T}{2}\right), \quad (\text{S18})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{T + P - R - S}{2}\right). \quad (\text{S19})$$

378 As before, SCT allows us to predict the mutual invasion of two strategy using the
 379 following inequality:

$$\rho < \frac{\kappa_j}{\kappa_i} < \frac{1}{\rho}. \quad (\text{S20})$$

380 Then, simple algebra reveals that each condition in mutual invasion inequality
 381 can be equivalently described by the payoff inequality:

$$\rho < \frac{\kappa_2}{\kappa_1} \iff R < T \quad (\text{S21})$$

$$\rho > \frac{\kappa_2}{\kappa_1} \iff R > T \quad (\text{S22})$$

$$\frac{\kappa_2}{\kappa_1} < \frac{1}{\rho} \iff P < S \quad (\text{S23})$$

$$\frac{\kappa_2}{\kappa_1} > \frac{1}{\rho} \iff P > S \quad (\text{S24})$$

382 These inequalities then allow us to classify outcomes of two-strategy games from the
 383 perspective of SCT. Note that the inequality between R and P does not affect the
 384 outcome.

385 **S3 Applications to mechanisms for the evolution of coopera-** 386 **tion**

387 We consider five mechanisms for the evolution of cooperation, all of which can be
 388 represented as a simple 2×2 payoff matrix (Nowak, 2006). Here, we go through each
 389 mechanism and derive measures for niche and fitness differences.

390 **Prisoner's dilemma.** Here, we consider a special case of Prisoner's dilemma
 391 which can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c \\ b & 0 \end{pmatrix}, \quad (\text{S25})$$

392 where b represents the benefit for the recipient and c represents cost for the donor.
 393 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0 \quad (\text{S26})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c). \quad (\text{S27})$$

394 Therefore, as long as the cost is greater than zero, prisoner's dilemma always results
 395 in the competitive exclusion of cooperation.

396 **Kin selection.** Kin selection can be described by the following payoff matrix:

$$\begin{pmatrix} (b - c)(1 + r) & br - c \\ b - rc & 0 \end{pmatrix}, \quad (\text{S28})$$

397 where r represents the genetic relatedness. In this case, the niche and fitness differ-
 398 ences correspond to:

$$1 - \rho = 0 \quad (\text{S29})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(c - br). \quad (\text{S30})$$

399 Therefore, defection will exclude cooperation when $c > br$. Equivalently, cooperation
 400 will be favored when $r > c/b$, which recovers the classic result.

401 **Direct reciprocity.** Direct reciprocity can be described by the following payoff
 402 matrix:

$$\begin{pmatrix} (b - c)/(1 - w) & -c \\ b & 0 \end{pmatrix}, \quad (\text{S31})$$

where w represents the probability of next round. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) \quad (\text{S32})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right). \quad (\text{S33})$$

Therefore, defection will exclude cooperation when

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S34})$$

$$\iff \exp\left(\frac{(b - c)/(1 - w) + c - b}{2}\right) < \exp\left(\frac{b + c - (b - c)/(1 - w)}{2}\right) \quad (\text{S35})$$

$$\iff bw < c \quad (\text{S36})$$

However, note that the following equality holds whenever $c > 0$:

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}. \quad (\text{S37})$$

Therefore, when $w > c/b$, we will have

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1} < \rho, \quad (\text{S38})$$

which implies priority effect (equivalently, bistability). Even as $w \rightarrow 1^-$, this inequality holds and therefore, cooperation cannot competitively exclude defection.

Indirect reciprocity. Indirect reciprocity can be described by the following payoff matrix:

$$\begin{pmatrix} b - c & -c(1 - q) \\ b(1 - q) & 0 \end{pmatrix}, \quad (\text{S39})$$

where q represents social acquaintanceship. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 1 - \exp\left(\frac{(b - c)q}{2}\right), \quad (\text{S40})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{2c - (b + c)q}{2}\right). \quad (\text{S41})$$

In this case, the condition for the dominance of defection (and therefore competitive exclusion of cooperation) can be written as:

$$\rho < \frac{\kappa_2}{\kappa_1} \quad (\text{S42})$$

$$\iff bq < c \quad (\text{S43})$$

416 Moreover, the following inequality always holds as long as $c > 0$.

$$\frac{1}{\rho} < \frac{\kappa_2}{\kappa_1}, \quad (\text{S44})$$

417 which implies that coexistence between cooperation and defection is not possible.
 418 Therefore, when $q < c/b$, then defection will competitively exclude cooperation.
 419 Otherwise, the system will enter the priority-effect regime.

420 Interestingly, when $q \rightarrow 1$, we have

$$\rho = \exp\left(\frac{b-c}{2}\right) \quad (\text{S45})$$

$$\frac{\kappa_2}{\kappa_1} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S46})$$

$$\frac{1}{\rho} = \exp\left(\frac{c-b}{2}\right) \quad (\text{S47})$$

$$(\text{S48})$$

421 In other words, the following inequality will hold:

$$\frac{1}{\rho} = \frac{\kappa_2}{\kappa_1} < \rho \quad (\text{S49})$$

422 **Network reciprocity.** Network reciprocity can be described by the following
 423 payoff matrix:

$$\begin{pmatrix} b-c & H-c \\ b-H & 0 \end{pmatrix}, \quad (\text{S50})$$

424 where $H = [(b-c)k - 2c]/[(k+1)(k-2)]$ and k represents the number of neighbors.
 425 In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S51})$$

$$\frac{\kappa_2}{\kappa_1} = \exp(-H + c), \quad (\text{S52})$$

426 which implies that there is complete niche overlap and that fitness difference decreases
 427 with H . Since H is a monotonically decreasing function of k when $k > 2$ and $b/c \geq 2$,
 428 fitness difference will increase with k . In particular, cooperation will competitively
 429 exclude defection when $H > c$, meaning that:

$$b/c > k. \quad (\text{S53})$$

430 **Group selection.** Group selection can be described by the following payoff
 431 matrix:

$$\begin{pmatrix} (b-c)(m+n) & (b-c)m - cn \\ bn & 0 \end{pmatrix}, \quad (\text{S54})$$

where m represents the number of groups and n represents the group size. In this case, the niche and fitness differences correspond to:

$$1 - \rho = 0, \quad (\text{S55})$$

$$\frac{\kappa_2}{\kappa_1} = \exp((c - b)m + cn), \quad (\text{S56})$$

which implies that there is complete niche overlap. The fitness difference will increase with n but decrease with m , provided that $b > c$. In particular, cooperation will competitively exclude defection under the following condition:

$$1 + \frac{n}{m} < \frac{b}{c}. \quad (\text{S57})$$

References

- Adler, P. B., M. Detto, S. P. Ellner, T. L. Gibbs, Z. J. Gold, S. J. Leonard, J. I. Levine, A. E. Schiffer, C. Song, M. Stemkovski, et al. (2026). What have we learned from empirical applications of modern coexistence theory? *Ecology letters* 29(6), e70404.
- Adler, P. B., J. HilleRisLambers, and J. M. Levine (2007). A niche for neutrality. *Ecology letters* 10(2), 95–104.
- Allesina, S. (2026). Global stability of ecological and evolutionary dynamics via equivalence. *Proceedings of the National Academy of Sciences* 123(13), e2534915123.
- Apicella, C. L. and J. B. Silk (2019). The evolution of human cooperation. *Current Biology* 29(11), R447–R450.
- Archetti, M. and I. Scheuring (2011). Coexistence of cooperation and defection in public goods games. *Evolution* 65(4), 1140–1148.
- Axelrod, R. and W. D. Hamilton (1981). The evolution of cooperation. *science* 211(4489), 1390–1396.
- Bodin, F., G. Wang, and J. B. Plotkin (2026). Eco-evolutionary games in noisy environments. *bioRxiv*, 2026–05.
- Carlson, M. and D. Botstein (1982). Two differentially regulated mRNAs with different 5' ends encode secreted and intracellular forms of yeast invertase. *Cell* 28(1), 145–154.
- Chesson, P. (2000). Mechanisms of maintenance of species diversity. *Annual review of Ecology and Systematics* 31(1), 343–366.

- 460 Clutton-Brock, T. (2009). Cooperation between non-kin in animal societies. *Nature* 462(7269), 51–57.
- 462 Cooney, D. B. (2020). Analysis of multilevel replicator dynamics for general two-
463 strategy social dilemma. *Bulletin of Mathematical Biology* 82(6), 66.
- 464 Cordero, O. X., L.-A. Ventouras, E. F. DeLong, and M. F. Polz (2012). Public good
465 dynamics drive evolution of iron acquisition strategies in natural bacterioplankton
466 populations. *Proceedings of the National Academy of Sciences* 109(49), 20059–
467 20064.
- 468 Cressman, R. (2003). *Evolutionary dynamics and extensive form games*, Volume 5.
469 MIT Press.
- 470 Cressman, R. and Y. Tao (2014). The replicator equation and other game dynamics.
471 *Proceedings of the National Academy of Sciences* 111(supplement_3), 10810–10817.
- 472 Dickinson, J. R. (1998). *Metabolism and molecular physiology of Saccharomyces*
473 *cerevisiae*. Taylor & Francis.
- 474 Doebeli, M. and C. Hauert (2005). Models of cooperation based on the Prisoner’s
475 Dilemma and the Snowdrift game. *Ecology letters* 8(7), 748–766.
- 476 Doebeli, M., C. Hauert, and T. Killingback (2004). The evolutionary origin of coop-
477 erators and defectors. *science* 306(5697), 859–862.
- 478 Fletcher, J. A. and M. Doebeli (2008). A simple and general explanation for the evolu-
479 tion of altruism. *Proceedings of the Royal Society B: Biological Sciences* 276(1654),
480 13.
- 481 Godoy, O., N. J. Kraft, and J. M. Levine (2014). Phylogenetic relatedness and the
482 determinants of competitive outcomes. *Ecology letters* 17(7), 836–844.
- 483 Godoy, O. and J. M. Levine (2014). Phenology effects on invasion success: insights
484 from coupling field experiments to coexistence theory. *Ecology* 95(3), 726–736.
- 485 Gokhale, C. S. and A. Traulsen (2025). Higher-order equivalence of Lotka-Volterra
486 and replicator dynamics. *bioRxiv*, 2025–03.
- 487 Gore, J., H. Youk, and A. Van Oudenaarden (2009). Snowdrift game dynamics and
488 facultative cheating in yeast. *Nature* 459(7244), 253–256.
- 489 Hamilton, W. D. (1964). The genetical evolution of social behaviour. II. *Journal of*
490 *theoretical biology* 7(1), 17–52.
- 491 Hauert, C. (2002). Effects of space in 2×2 games. *International Journal of Bifur-*
492 *cation and Chaos* 12(07), 1531–1548.

- 493 Hauert, C. and M. Doebeli (2004). Spatial structure often inhibits the evolution of
494 cooperation in the snowdrift game. *Nature* 428(6983), 643–646.
- 495 Hilbe, C., Š. Šimsa, K. Chatterjee, and M. A. Nowak (2018). Evolution of cooperation
496 in stochastic games. *Nature* 559(7713), 246–249.
- 497 Hofbauer, J. (1981). On the occurrence of limit cycles in the volterra-lotka equation.
498 *Nonlinear Analysis: Theory, Methods & Applications* 5(9), 1003–1007.
- 499 Hofbauer, J. and K. Sigmund (1998). *Evolutionary games and population dynamics*.
500 Cambridge university press.
- 501 Kingma, S. A., P. Santema, M. Taborsky, and J. Komdeur (2014). Group augmen-
502 tation and the evolution of cooperation. *Trends in ecology & evolution* 29(8),
503 476–484.
- 504 Kocher, S. D., R. Mallarino, B. E. Rubin, D. W. Yu, H. E. Hoekstra, and N. E.
505 Pierce (2018). The genetic basis of a social polymorphism in halictid bees. *Nature*
506 *Communications* 9(1), 4338.
- 507 Kraft, N. J., O. Godoy, and J. M. Levine (2015). Plant functional traits and the mul-
508 tidimensional nature of species coexistence. *Proceedings of the National Academy*
509 *of Sciences* 112(3), 797–802.
- 510 Lee, I. P. A., O. T. Eldakar, J. P. Gogarten, and C. P. Andam (2022). Bacterial co-
511 operation through horizontal gene transfer. *Trends in Ecology & Evolution* 37(3),
512 223–232.
- 513 Lewontin, R. C. (1961). Evolution and the theory of games. *Journal of Theoretical*
514 *Biology* 1(3), 382–403.
- 515 Lieberman, E., C. Hauert, and M. A. Nowak (2005). Evolutionary dynamics on
516 graphs. *Nature* 433(7023), 312–316.
- 517 Lin, H., D. Wang, Q. Wang, J. Mao, Y. Bai, and J. Qu (2024). Interspecific compe-
518 tition prevents the proliferation of social cheaters in an unstructured environment.
519 *The ISME Journal* 18(1), wrad038.
- 520 May, R. M. (1987). More evolution of cooperation. *Nature* 327(6117), 15–17.
- 521 Michel-Mata, S., M. Kawakatsu, J. Sartini, T. A. Kessinger, J. B. Plotkin, and C. E.
522 Tarnita (2024). The evolution of private reputations in information-abundant
523 landscapes. *Nature* 634(8035), 883–889.
- 524 Motro, U. (1991). Co-operation and defection: playing the field and the ESS. *Journal*
525 *of Theoretical Biology* 151(2), 145–154.

526 Nowak, M. A. (2006). Five rules for the evolution of cooperation. *science* 314(5805),
527 1560–1563.

528 Nowak, M. A. and R. M. May (1992). Evolutionary games and spatial chaos. *na-*
529 *ture* 359(6398), 826–829.

530 Nowak, M. A. and K. Sigmund (1998). Evolution of indirect reciprocity by image
531 scoring. *Nature* 393(6685), 573–577.

532 Nowak, M. A. and K. Sigmund (2005). Evolution of indirect reciprocity. *Na-*
533 *ture* 437(7063), 1291–1298.

534 Nowak, M. A., C. E. Tarnita, and T. Antal (2010). Evolutionary dynamics in struc-
535 tured populations. *Philosophical Transactions of the Royal Society B: Biological*
536 *Sciences* 365(1537), 19.

537 Page, K. M. and M. A. Nowak (2002). Unifying evolutionary dynamics. *Journal of*
538 *theoretical biology* 219(1), 93–98.

539 Palmer, J. D. and K. R. Foster (2022). Bacterial species rarely work together. *Sci-*
540 *ence* 376(6593), 581–582.

541 Park, S. W., S. Cobey, C. J. E. Metcalf, J. M. Levine, and B. T. Grenfell (2024).
542 Predicting pathogen mutual invasibility and co-circulation. *Science* 386(6718),
543 175–179.

544 Park, S. W., J. Levine, and B. Grenfell (2026). Unifying coexistence theory for
545 ecological communities and pathogen strain competition. *bioRxiv*, 2026–05.

546 Pollak, S., M. Gralka, Y. Sato, J. Schwartzman, L. Lu, and O. X. Cordero (2021).
547 Public good exploitation in natural bacterioplankton communities. *Science Ad-*
548 *vances* 7(31), eabi4717.

549 Rakoff-Nahoum, S., K. R. Foster, and L. E. Comstock (2016). The evolution of
550 cooperation within the gut microbiota. *Nature* 533(7602), 255–259.

551 Schuster, P. and K. Sigmund (1983). Replicator dynamics. *Journal of theoretical*
552 *biology* 100(3), 533–538.

553 Sieben, A. J., J. R. Mihaljevic, and L. G. Shoemaker (2022). Quantifying mechanisms
554 of coexistence in disease ecology. *Ecology* 103(12), e3819.

555 Slobodkin, L. B. (1964). The strategy of evolution. *American scientist* 52(3), 342–
556 357.

557 Smith, J. M. and G. R. Price (1973). The logic of animal conflict. *Nature* 246(5427),
558 15–18.

- 559 Souza, M. O., J. M. Pacheco, and F. C. Santos (2009). Evolution of cooperation
560 under N-person snowdrift games. *Journal of Theoretical Biology* 260(4), 581–588.
- 561 Steidinger, B. S. and J. D. Bever (2014). The coexistence of hosts with different
562 abilities to discriminate against cheater partners: an evolutionary game-theory
563 approach. *The American Naturalist* 183(6), 762–770.
- 564 Szabó, G. and G. Fath (2007). Evolutionary games on graphs. *Physics reports* 446(4-
565 6), 97–216.
- 566 Tarnita, C. E. and A. Traulsen (2025). Reconciling ecology and evolutionary game
567 theory or “When not to think cooperation”. *Proceedings of the National Academy*
568 *of Sciences* 122(14), e2413847122.
- 569 Taylor, P. D. and L. B. Jonker (1978). Evolutionary stable strategies and game
570 dynamics. *Mathematical biosciences* 40(1-2), 145–156.
- 571 Traulsen, A. and M. A. Nowak (2006). Evolution of cooperation by multilevel selec-
572 tion. *Proceedings of the National Academy of Sciences* 103(29), 10952–10955.
- 573 Trivers, R. L. (1971). The evolution of reciprocal altruism. *The Quarterly review of*
574 *biology* 46(1), 35–57.
- 575 Wang, G., Q. Su, L. Wang, and J. B. Plotkin (2025). Evolution of social behaviors
576 in noisy environments. *arXiv*.
- 577 Wedekind, C. and M. Milinski (2000). Cooperation through image scoring in humans.
578 *Science* 288(5467), 850–852.
- 579 West, S. A., G. A. Cooper, M. B. Ghoul, and A. S. Griffin (2021). Ten recent insights
580 for our understanding of cooperation. *Nature ecology & evolution* 5(4), 419–430.
- 581 West, S. A., A. S. Griffin, and A. Gardner (2007). Evolutionary explanations for
582 cooperation. *Current biology* 17(16), R661–R672.